BOARD OF SECONDARY EDUCATION, KARACHI

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

SECTION "A" MULTIPLE CHOICE QUESTIONS (MCQs)

Maximum Marks: 50

Note: NOTE: (i) Attempt all the Questions. (ii) Write only the answer against the proper number of the question and its part according to question paper. (iii) Each question carries 2 Marks

Q1. Choose the correct Answer from the options given below for each question:

Q1. {0,1,2,3,.....} is the set of:

(A) Integers
(B) Whole-Numbers
(C) Even Numbers
(D) Prime Numbers

Q2. $(-6, -8)$ is in quadrant:

(A) 1st
(B) 2nd
(C) 3rd
(D) 4th

Q3. If log_a 16 = 4 than a =:

(A) 3
(B) 4
(C) 2
(D) 16

Q4. If $a + b = 2$ and $a - b = 2$ then $a^2 + b^2$ =:

(A) 2
(B) −2
(C) −1
(D) 4

Q5. $(\sqrt{x} + \sqrt{y})(\sqrt{x} - \sqrt{y})$:

(A) $x + y$
(B) $x^2 + y^2$
(C) $x - y$
(D) $x^2 - y^2$

Q6. The solution set of $\sqrt{(y-2)} = -4$ is:

(A) {18}
(B) {±4}
(C) {}
(D) None of these

Q7. Order of [a + b c] is

(A) 1 × 2 (B) 3 × 1

(C) 2 × 3 (D) 1 × 3

Q8. The degree of polynomial $x^3y + xy + 25$ is

(A) 3 (B) 5

(C) 2 (D) 4

Q9. A triangle having no side congruent, is called:

(A) Equilateral Triangle (B) Isosceles Triangle

(C) Scalene Triangle (D) None of these

Q10. The value of sec 45° is

(A) 1/2 (B) √2

(C) √3 (D) 1/√2

Q11. If R = {(1,2), (2,3), (3,4)} then Dom(R) = :

(A) {1} (B) {1,2}

(C) {2,3,4} (D) {1,2,3}

Q12. Additive inverse of a − b is:

(A) 1/(a−b) (B) a + b

(C) −a − b (D) b − a

Q13. The natural logarithm has the base:

(A) π (B) e

(C) 10 (D) 0

Q14. $(x^2 - y^2) \div (x - y) =$:

(A) x + y (B) $x^2 + y^2$

(C) x − y (D) None of these

Q15. H. C. F. of $x^3 - 8$ and $x^4 - 16$ is:

(A) $x^3 - 8$ (B) $x^4 - 16$

(C) $x^2 - x + 2$ (D) None of these

Q16. $2\sqrt{8} + 6\sqrt{8} - 2\sqrt{2} =$:

(A) 6√2 (B) 4√2

(C) 12√2 (D) 10√2

Q17. If [[2, 2], [8, p]] is a singular matrix then the value of p is:

(A) 3 (B) 5

(C) 6 (D) 7

Q18. A chord which passes through the centre of the circle, is called:

(A) diameter (B) radius

(C) tangent (D) none of these

Q19. cos 80° =:

(A) Sec 10° (B) Sin 10°

(C) Sin 80° (D) Cot 10°

Q20. The characteristic of log 0.0226 is:

(A) 2 (B) −2

(C) $\bar{1}$ (D) 1

Q21. L.C.M. of $9x^2$ and $15x^3$ is:

(A) $45x^3$ (B) $3x^2$

(C) $3x^3$ (D) $15x^3$

Q22. cotθ / cosθ =:

(A) cos θ (B) sin θ

(C) cosec θ (D) sec θ

Q23. $x^4 + 16x^2$ will be a perfect square by adding:

(A) 8 (B) 16

(C) 64 (D) 4

Q24. The sum of two complementary angles is:

(A) 360° (B) 90°

(C) 180° (D) 45°

Q25. $Cosec^2\theta - 1 =$:

(A) $Cos^2\theta$ (B) $Sin^2\theta$

(C) $Cot^2\theta$ (D) $Sec^2\theta$

SECTION “B” (Short Answer Question)

Maximum Marks: 30

Note: NOTE: Attempt any SIX parts from this section, Selecting ONE part from each question. All question carry equal marks.

Q2 (i). If A = {1,2,3,4}, B = {2,4,6,8} then prove that $A\Delta B = (A - B) \cup (B - A)$

Q2 (ii). If $a = \sqrt{10} + 3$ then find the value of $a^2 + 1/a^2$

Q3 (i). For What values of p and q $4x^4 + 12x^3 + 25x^2 + px + q$ will be a perfect square?

Q3 (ii). Find the value of 0.87 / ((28.9)(0.785)) with the help of logarithm table.

Q4 (i). Find the value of $x - y$ when $x + y = -9$ and $xy = 20$

Q4 (ii). Find the solution set of $\sqrt{(25y - 6)} = 4\sqrt{(y + 3)}$ and also verify your answer.

Q5 (i). Solve the equations $2x - y = -2$; $x + 2y = 3$ with the help of Cramer's rule.

Q5 (ii). Eliminate "t" from $x = a(1-t^2)/(1+t^2)$, $y = b(1-t^2)/(2t^2)$

Q6 (i). Prove that $\cot\beta + \tan\beta = \cot\beta + \sec^2\beta$ (The above Question is misprinted in board's paper, the correct question is given below)

Q6 (i). Prove that $\cot\beta + \tan\beta = \cot\beta \sec^2\beta$

Q6 (ii). Find the solution set of $-6 + |5x - 3| = 3$

Q7 (i). A kite has 120m of string attached to it. When at an angle of elevation of 60°. How far is it above the hand holding it?

Q7 (ii). Solve $2x^2 + 5x - 3 = 0$ by using quadratic formula.

Section C (Detailed –Answer Question)

Maximum Marks: 20

> Note: NOTE: Attempt any TWO questions from this section including Question No. 8 which is compulsory.

Q8. If two lines intersect, the vertically opposite angles so formed are congruent. Prove it.

Q9. Factorize the following: (Any four)

(i) $x^{12} - y^{12}$

(ii) $36x^2 + 154x - 36$

(iii) $1 - x^2 - y^2 + 2xy$

(iv) $4a^4 + 625b^4$

(v) $x^3 - x^2 + 2$

(vi) $a^6 - 27b^3 - 8c^6 - 18a^2bc^2$

Q10. Construct a triangle ABC in which mAB = mBC = 6cm and m∠B = 50° and also write steps of construction.

Q11. If a transversal intersects two coplanar lines such that the pair of alternate angles are congruent, then the lines are parallel. Prove it.